

Spline-SVP v1: Reference Tests and Gaussian-Null Calibration

1 Objective

The objective of this stage was to validate the exhaustive spline-SVP v1 reference implementation and to calibrate the local validity statistic under the Gaussian affine null model.

The work had two main components:

1. numerical tests of the validity statistic, fixed-partition fitting, partition enumeration, and lexicographic oracle;
2. Monte Carlo calibration of the validity threshold γ , including its dependence on interval length and preliminary sensitivity to the design geometry.

The mathematical definition of spline-SVP v1 and the derivation of the segment fitting cost were given previously and are not repeated here.

Throughout this note,

$$t_1 < \dots < t_n$$

denote the observation locations, while

$$x_1, \dots, x_n$$

denote the corresponding observed values. The symbols $i, j, \dots$ denote observation indices. The symbols $s < t$ denote the endpoints of a generic candidate interval, and $r \in (s, t)$ denotes a candidate internal knot location.

2 Reference-implementation tests

For a candidate interval $(s, t]$, the current state-independent validity statistic is

$$T_{s,t} = \frac{RSS_0(s, t) - \min_{s < r < t} RSS_1(s, r, t)}{\sigma^2},$$

where

$$RSS_0(s, t) = \min_{a, b \in \mathbb{R}} \sum_{i: s < t_i \leq t} (x_i - a - bt_i)^2$$

is the residual sum of squares under a single affine fit, and

$$RSS_1(s, r, t) = \min_{a, b, c \in \mathbb{R}} \sum_{i: s < t_i \leq t} [x_i - a - bt_i - c(t_i - r)_+]^2$$

is the residual sum of squares under a continuous one-knot alternative.

Several controlled cases were first used to test the statistic.

For very short intervals, the current v1 convention gives

$$T_{s,t} = 0$$

when the interval contains at most two observations. This behaviour was verified numerically.

For an exact affine signal,

$$x_i = 2 + 3t_i,$$

the computed value was

$$T = 3.23 \times 10^{-29},$$

which is zero up to floating-point precision.

For an exact continuous signal with a strong slope change, the statistic was

$$T = 30,$$

showing the expected large improvement from introducing an internal knot.

Scale invariance was checked by multiplying both the observations and the noise standard deviation by 7. The values before and after scaling were both

$$T = 37.89597.$$

The nested-model property was also checked numerically. On one test example,

$$RSS_0 = 11.62365, \quad \min_r RSS_1 = 2.149657,$$

with the best candidate knot at

$$r = 6.$$

The remaining components of the exhaustive reference implementation were tested separately. The results are summarized in Table 1.

Test	Result
Fixed-partition fit	For a one-segment fit, the spline cost 0.6697334 agreed with the ordinary affine-regression RSS 0.6697334; absolute difference 1.22×10^{-15} .
Shared-state implementation	For the test partition (1, 5, 8, 10), adjacent segment formulas gave identical fitted values at each common boundary; maximum discrepancy 0.
Partition enumeration	For every $n = 2, \dots, 10$, exactly 2^{n-2} distinct candidate partitions were generated.
Partition validity	With $\gamma = 4$, the one-segment partition (1, 10) was rejected since $T_{1,10} = 30$, while (1, 5, 10) was accepted since both segment statistics were numerically zero.
Lexicographic selection	Explicit minimization over the table of valid partitions gave $K^* = 2$ and minimum cost 0.3022311, matching the oracle.
End-to-end consistency	Both explicit lexicographic ordering and the oracle selected $\hat{\tau} = (1, 5, 10)$.

Table 1: Principal tests of the spline-SVP v1 exhaustive reference implementation.

The shared-state check is an implementation test rather than a test of continuity as a mathematical property. Continuity is already guaranteed by the parameterization because adjacent segments use the same fitted boundary state.

3 Gaussian-null calibration

For a candidate interval $(s, t]$, consider the null hypothesis

$$H_0^{s,t} : \quad X_i = a + bt_i + \varepsilon_i, \quad s < t_i \leq t,$$

where

$$\varepsilon_i \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma^2).$$

Under $H_0^{s,t}$, the mean signal is affine on the interval and there is no internal slope changepoint.

For a prescribed local significance level α , the calibration problem is to choose a threshold γ such that

$$\Pr_{H_0}(T_{s,t} > \gamma) \approx \alpha.$$

Equivalently, γ is the $(1 - \alpha)$ -quantile of the null distribution of $T_{s,t}$.

Adding an affine function of t to the observations does not change the statistic because both the null and alternative models contain an affine component. Likewise, multiplying both x_i and σ by the same constant leaves $T_{s,t}$ unchanged.

These invariances were checked numerically. Adding the affine trend

$$7 - 2.3t$$

changed the statistic by only

$$2.31 \times 10^{-14},$$

and subsequently multiplying both the observations and σ by 4.7 changed it by

$$7.11 \times 10^{-15}.$$

Therefore the null calibration was performed using the standardized model

$$a = 0, \quad b = 0, \quad \sigma = 1.$$

3.1 Dependence on interval length

Let

$$m = \#\{i : s < t_i \leq t\}$$

denote the number of observations entering the local validity statistic.

For each

$$m \in \{3, 4, 5, 6, 8, 10, 15, 20, 30, 40, 50\},$$

$B = 10\,000$ independent Gaussian-null datasets were generated on an equally spaced design. For each simulated dataset, $T_{s,t}$ was computed and the empirical 90%, 95%, and 99% quantiles were estimated.

m	$q_{0.90}$	$q_{0.95}$	$q_{0.99}$
3	2.734	3.989	6.801
4	3.484	4.809	7.645
5	3.924	5.250	8.167
6	4.257	5.528	8.399
8	4.689	6.007	9.019
10	4.883	6.301	9.756
15	5.134	6.435	9.641
20	5.481	6.886	10.272
30	5.740	7.064	10.121
40	5.799	7.235	10.148
50	5.998	7.378	10.637

Table 2: Empirical Gaussian-null quantiles using $B = 10\,000$ simulations for each interval length.

The dependence on m is substantial. For example,

$$q_{0.95}(3) = 3.989, \quad q_{0.95}(10) = 6.301, \quad q_{0.95}(50) = 7.378.$$

Thus, for equally spaced designs, a fixed threshold γ does not maintain a constant local type-I error across interval lengths.

The calibrated validity rule should therefore take the form

$$T_{s,t} \leq \gamma_{m,\alpha},$$

where

$$\gamma_{m,\alpha} = F_m^{-1}(1 - \alpha)$$

and F_m denotes the null distribution function of T for an interval containing m observations.

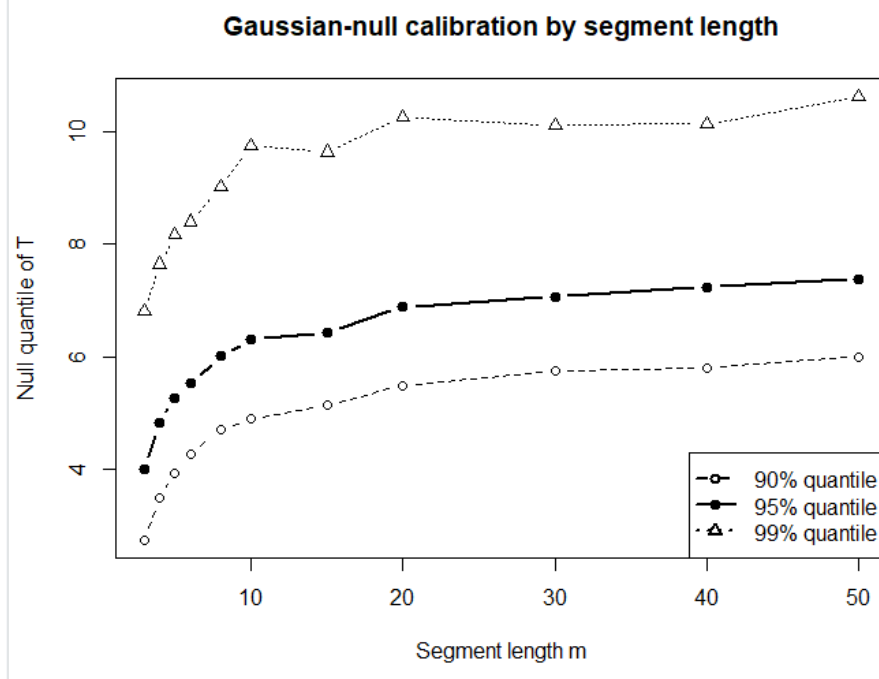


Figure 1: Empirical 90%, 95%, and 99% Gaussian-null quantiles of T as functions of interval length.

3.2 Validation of the calibrated threshold

For

$$m = 20, \quad \alpha = 0.05,$$

a calibration sample of 20 000 independent null realizations gave the empirical threshold

$$\hat{\gamma}_{20,0.05} = 6.820359.$$

This threshold was then applied to a new independent sample of 20 000 null realizations. The observed rejection probability was

$$0.0495,$$

which is close to the target level

$$0.05.$$

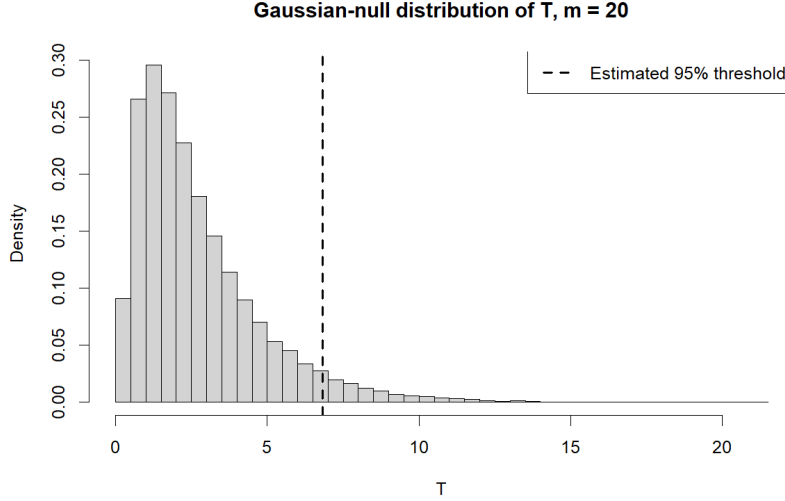


Figure 2: Empirical null distribution of T for $m = 20$. The dashed line is the estimated 95% threshold.

4 Preliminary design-geometry sensitivity

The preceding calibration used equally spaced observation locations. Since the one-knot regressors depend on the relative positions of the t_i , the null distribution may also depend on the local design geometry.

To investigate this, the number of observations was fixed at

$$m = 20,$$

and four deterministic designs were compared using $B = 10\,000$ simulations per design.

Design	$q_{0.95}$	Rejection using equal-design threshold
Equally spaced	6.837	0.0500
Alternating gaps	6.818	0.0493
Increasing gaps	6.920	0.0519
Clustered	7.188	0.0593

Table 3: Preliminary design-sensitivity results for $m = 20$. The rejection probabilities use the 95% threshold calibrated under the equally spaced design.

The first three designs give similar null behavior. The clustered design produces a somewhat larger 95% quantile and a rejection probability of

$$0.0593$$

when the equally spaced threshold is reused.

Thus the current experiment suggests that design geometry can affect the null distribution, although the observed effect is moderate for the configurations considered here.

For equally spaced designs, the present calibration supports the rule

$$\gamma = \gamma_{m,\alpha}.$$

For general irregular designs, a more design-specific threshold may be required.

5 Week 4 status

The current exhaustive spline-SVP implementation has now been tested through componentwise and end-to-end consistency checks. The local validity statistic has also been calibrated under the Gaussian affine null for equally spaced designs.

The main empirical result is that the null distribution of $T_{s,t}$ changes with the number of observations in the tested interval. Consequently, the provisional fixed threshold used in the first synthetic example should be replaced, for a prescribed local level α , by a calibrated threshold

$$\gamma_{m,\alpha}.$$

Preliminary experiments also indicate some sensitivity to irregular design geometry. A more complete treatment of this dependence remains an open question.

The next algorithmic stage is to use the exhaustive implementation as a correctness oracle while developing the functional dynamic-programming recurrence.